A Conversational Agent for Rare Eye Disease Knowledge Support in Local Italian Healthcare Contexts: The ELENA Framework

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***Abstract*—Accessing global medical knowledge on rare ophthalmic diseases remains a challenge for small healthcare centers, as relevant literature and case reports are scattered across multiple languages and formats. To bridge this gap, we introduce ELENA (Eye-disease Language-model for Evidence-based Navigation and Assistance), a retrieval-augmented chatbot designed to provide multilingual access to specialized medical knowledge in ophthalmic rare diseases. ELENA combines a quantized Large Language model with a lightweight retrieval-augmented generation pipeline, where all source documents are pre-translated into English to ensure consistent embedding and retrieval. Italian user queries are automatically translated, processed through the pipeline, and the generated responses are translated back into Italian, yielding fluent and contextually grounded answers. Experiments on a curated corpus of ophthalmic rare-disease documents confirm the technical feasibility of deploying such systems even on modest hardware, while a graphical interface ensures accessibility for non-technical clinical staff. ELENA thus demonstrates how an efficient, privacy-preserving, and human-friendly AI assistant can democratize access to international medical knowledge within local healthcare environments.**

***Index Terms*—Retrieval-Augmented Generation, Large Language Models, Ophthalmology, Rare Diseases, Local Healthcare Institutions**

I. Introduction

In recent years, the integration of Artificial Intelligence (AI) into clinical and research workflows has shown remarkable potential in supporting medical professionals across a wide range of specialties. However, within the Italian healthcare system, small local health authorities (Aziende Sanitarie Locali, ASL) often face substantial limitations in terms of financial resources, technological infrastructure, and specialized personnel. This issue becomes even more pronounced when dealing with a case of a rare disease, where data scarcity and fragmented literature hinder effective diagnosis and research progress. In the ophthalmic domain, rare diseases have been estimated to have a prevalence of approximately 6-8%, with the majority being genetic in origin [1]. According to the National Eye Institute, over 900 rare eye diseases are listed, with one common form, retinitis pigmentosa, affecting an estimated 1 in 5,000 people. On average, it takes 8 years to receive an accurate diagnosis for a rare disease, and about 95% of them lack an FDA-approved treatment [2]. In previous work, efforts were made in this direction to develop intelligent systems for medical diagnostic support aimed at promptly identifying the presence of ocular diseases [3]–[5]. In this scenario, it becomes crucial to streamline the diagnostic process by ensuring that even small research units and local healthcare institutions have timely access to the most up-to-date medical knowledge. To address these issues, this work introduces ELENA (Eye-disease Language-model for Evidence-based Navigation and Assistance), a retrieval-augmented conversational framework specifically designed to assist clinicians and young medical researchers in exploring rare ophthalmic diseases. The framework combines large language models (LLMs) with retrieval-augmented generation (RAG) to provide accurate, context-aware, and human-like responses to user queries posed in natural Italian language (Fig.1), considering the increasing role of LLMs in healthcare [6], [7].

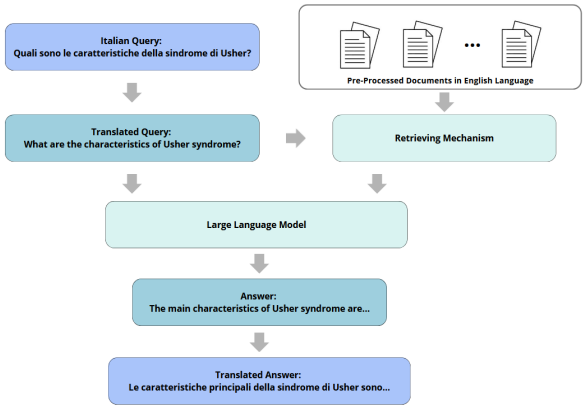


Fig. 1. ELENA framework conceptual architecture.

ELENA operates by translating Italian user questions into English, performing semantic retrieval across a multilingual document repository, and generating coherent answers that are subsequently translated back into Italian. In this way, even

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smaller organizations can access documents in other languages and up-to-date information on disease treatments, including most rare ones. This architecture ensures that the system leverages the strengths of English-based medical corpora while maintaining accessibility for Italian-speaking healthcare professionals. The framework is designed as a scalable, privacy-preserving tool that can be locally deployed in small healthcare institutions, helping democratize AI-driven medical support while safeguarding patient data.

II. Related Works

The concept of developing a chatbot leveraging a LLM to provide answers related to ophthalmology field has been explored in the literature. In [8], a cross-sectional study compared ChatGPT-generated responses to eye care questions with answers written by American Academy of Ophthalmology–affiliated ophthalmologists. Using 200 deidentified patient questions from an online forum, a panel of board-certified ophthalmologists assessed whether answers were AI- or human-generated, and evaluated accuracy, appropriateness, and potential for harm. The study found that ChatGPT responses were largely comparable to human-written answers in terms of correctness, likelihood of harm, and alignment with community standards. Considering solutions or studies that have focused on specific disorders, [9] and [10]. The first work evaluated an LLM chatbot (GPT-4) in ophthalmology, comparing its diagnostic accuracy and completeness to fellowship-trained glaucoma and retina specialists using both commonly asked questions and real patient cases. For glaucoma-related scenarios, the chatbot achieved higher mean ranks for accuracy and completeness compared with specialists. For retina-related scenarios, the chatbot matched specialists in accuracy and outperformed them in completeness. The second work introduced CataractBot, an LLM-based system designed to answer cataract surgery–related questions using a curated, expert-verified knowledge base. Deployed in a real-world study with patients, doctors, and coordinators, the bot demonstrated potential for providing instant, accessible, and privacy-preserving information. It supported multilingual and multimodal interactions, accommodated varying literacy levels, and helped mitigate power imbalances. However, an issue that emerges from the analysis of the state of the art is the near absence of studies addressing the domain of rare diseases. As discussed in [11], even GPT-4 shows poorer performance in the context of rare diseases. Furthermore, the literature includes studies proposing generic solutions for rare diseases in general, as in the case of [12], or, as observed in the works reviewed, focusing on a single type of disease within a specific domain. The proposed ELENA framework uniquely combines RAG with multilingual translation to support small Italian healthcare institutions, enabling questions posed in Italian to access a curated global knowledge base of rare eye disease documents. Unlike large, general-purpose LLMs, which may have broader but less domain-focused knowledge, ELENA relies exclusively on selected and regularly updated documents (also internal, considering the federated institution-oriented nature of the framework [13]). This strategy enhances domain-specific expertise while covering the full spectrum of rare eye diseases, rather than concentrating on a single condition, thereby ensuring both depth and breadth in the clinical knowledge accessible to users.

III. Methodology

The ELENA framework was designed to provide an effective multilingual conversational assistant for rare ophthalmic diseases within local Italian healthcare institutions. The system addresses one of the major challenges in real-world medical information retrieval: the coexistence of heterogeneous textual sources written in multiple languages. To ensure a unified semantic space for knowledge access, all source documents—initially collected in Italian, English, and other languages are pre-processed and automatically translated into English using a neural machine translation model. This step allows ELENA to operate within a consistent embedding space, facilitating accurate retrieval regardless of the document's original language. When a user poses a question in Italian, the query is first translated into English, embedded, and compared against the pre-processed corpus to retrieve the most semantically relevant passages. These retrieved segments form the contextual basis for the subsequent RAG phase, in which the LLM synthesizes an informed and human-like answer in English. Finally, the generated answer is translated back into Italian and presented to the user in a fluent, natural-language format. This multilingual pipeline ensures both linguistic accessibility for Italian healthcare professionals and semantic fidelity to the underlying clinical knowledge base, making ELENA a practical and privacy-preserving conversational tool for small-scale medical institutions.

A. Datasets

For the ophthalmology-focused implementation of the ELENA framework, a domain-specific dataset was constructed to evaluate the system's capacity to retrieve and synthesize information on rare eye diseases. In this conceptual stage of the frame, the dataset consists of a collection of plain-text (.txt) files, each dedicated to a distinct condition, including but not limited to: Retinitis pigmentosa, Leber congenital amaurosis, Usher syndrome, Juvenile macular degeneration, Leber hereditary optic neuropathy, Congenital optic atrophy, Fuchs' endothelial dystrophy, Hereditary neurotrophic keratitis, Reis-Bücklers corneal dystrophy, Congenital hereditary cataract, Microspherophakia, Marfan syndrome (ocular manifestations), Lowe syndrome, Alport syndrome, Parry-Romberg syndrome, Congenital aniridia, and Axenfeld-Rieger syndrome. Each document contains descriptive medical knowledge about the associated pathology, encompassing aspects such as clinical presentation, genetic etiology, diagnostic markers, and management approaches. While the current experimental setup relies on manually curated text files to ensure data consistency and traceability, the framework is agnostic to input format. In real-world deployments, ELENA could seamlessly incorporate

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heterogeneous data sources—including PDF reports, HTML-based medical repositories, or structured clinical notes—as long as they can be transformed into textual input for the retrieval engine. Furthermore, in a realistic operational scenario, the retrieval process should be conducted online to guarantee up-to-date information, with strict reliability criteria defined to ensure the trustworthiness of the accessed sources. This dataset thus serves as a controlled testbed for validating the feasibility and precision of retrieval-augmented reasoning within the ophthalmic rare-disease domain, providing a foundation for future scaling to institutional and multilingual data collections. Furthermore, to assess the validity of the generated outputs, the framework was trained and evaluated using the ReDis dataset (https://huggingface.co/datasets/guan-wang/ReDis-QA), which comprises 1,360 multiple-choice questions covering 205 distinct rare diseases. The questions are annotated according to disease properties such as symptoms, causes, effects, related disorders, and diagnosis, providing a structured framework for model assessment. However, the dataset exhibits some limitations: its modest size may be insufficient for training complex models from scratch; the coverage, although spanning 205 diseases, leaves many rare conditions unrepresented; and the multiple-choice format restricts generative, free-form QA capabilities. These limitations highlight the importance of complementing static datasets with dynamic retrieval from up-to-date scientific publications and official medical resources, ensuring that AI models can access the most recent and comprehensive knowledge without being constrained by a unique and, potentially outdated, dataset. However, the use of this dataset was crucial, owing to the presence of reference answers, which made possible to quantitatively evaluate the effectiveness of the proposed pipeline by measuring how closely the model's generated responses aligned with the medically expected answers.

B. Retrieval-Augmented Generation and Response Pipeline

In this study, we utilized Falcon-7B-Instruct [14], a 7-billion parameter instruction-tuned LLM designed for natural language understanding and generation. Falcon-7B-Instruct is based on the Falcon architecture, which has demonstrated strong performance in instruction-following tasks, while remaining computationally efficient compared to other models of similar scale. To enable deployment on hardware with limited memory, we employed 4-bit quantization which allowed the Falcon-7B-Instruct model to run efficiently on standard GPUs while maintaining high-quality generative capabilities, mimicking the modest hardware availability that would characterize a real deployment scenario. The modular design of the framework ensures that the underlying language model can be substituted with more advanced or larger-scale alternatives as these emerge, without necessitating major modifications to the retrieval or response-generation components. To enhance the accuracy and relevance of the model's responses, we implemented a RAG system. In this framework, the language model does not rely solely on its pre-trained knowledge but can access to external, updated documents on eye rare disease domain. Since the corpus consists of documents written in multiple languages, all texts were first translated into English to create a unified semantic space for retrieval and generation. To facilitate efficient retrieval, the input documents were segmented into smaller, semantically coherent chunks using a *RecursiveCharacterTextSplitter* with a chunk size of 800 characters and an overlap of 100. Each chunk was then embedded using the compact *all-MiniLM-L6-v2* [15] model from the Sentence-Transformers family, which provides lightweight yet semantically rich vector representations. The embeddings were stored in a vector database to enable similarity-based retrieval through approximate nearest neighbor search. When a user poses a question in Italian, the system first translates it into English, then it is used to retrieve the top-k most relevant document chunks. The retrieved information is concatenated into a contextual prompt and passed to the instruction-tuned LLM model to generate a coherent and medically accurate response in English. The generation prompt explicitly guides the model to produce a concise, human-like explanation of the queried disease or concept, summarizing key clinical aspects in three to five sentences. The prompt used is proposed in Table I.

This architecture thus bridges linguistic and computational gaps by combining cross-lingual translation, dense semantic retrieval, and instruction-based text generation, providing a scalable and privacy-preserving assistant for rare ophthalmologic disease management.

IV. Results

All experiments were conducted on Google Colab using a T4 GPU (15 GB RAM), simulating the limited computational resources typical of possible real-world clinical settings. Despite the relatively modest hardware resources, the use of 4-bit quantization allowed us to load and run the Falcon-7B-Instruct model efficiently, significantly reducing memory requirements without substantial loss in performance.

This setup enabled the execution of both the RAG pipeline and the LLM inference in a single environment. The system was able to process questions, perform semantic retrieval from the knowledge base, generate responses in English, and translate them back into Italian in real time. Given that the current study is primarily conceptual, the documents used for preliminary human testing of the framework were stored and accessed locally. The quality of the generated answers was evaluated using multiple metrics, including cosine similarity of embeddings, cosine similarity of TF-IDF vectors, BLEU, ROUGE-L, Levenshtein distance, and an aggregated similarity score. The results, summarized in Table II, show that the framework produces contextually relevant and semantically coherent answers, despite the limited computational resources.

Examining the results numerically, we can observe that the average values of the metrics are relatively low: the mean cosine similarity of embeddings is 0.379, the cosine similarity based on TF-IDF is 0.122, BLEU is 0.002, ROUGE-L is 0.036, and Levenshtein distance normalized similarity is 0.048. Even the combined mean similarity metric, which averages

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TABLE I
STRUCTURE OF THE GENERATION PROMPT USED IN THE ELENA FRAMEWORK.

Component	Content
System Role	*You are a medical expert in rare diseases.*
Instruction	*Use the information below to answer the user's question clearly and naturally.*
User Question	Italian question translated into English.
Relevant Information	Top-k retrieved document excerpts.
Task Directive	*Now write a concise, human-like answer that explains the disease or concept directly. If relevant, summarize the key features or causes of the condition in 3–5 sentences.*

TABLE II
EVALUATION METRICS FOR THE GENERATED ANSWERS.

Metric	Mean	Min	Max	Std
Cosine similarity (embedding)	0.379	0.094	0.732	0.151
Cosine similarity (TF-IDF)	0.122	0.000	0.511	0.150
BLEU	0.002	0.000	0.007	0.002
ROUGE-L	0.036	0.000	0.107	0.034
Levenshtein	0.048	0.013	0.139	0.031
Aggregated similarity	0.117	0.024	0.255	0.058

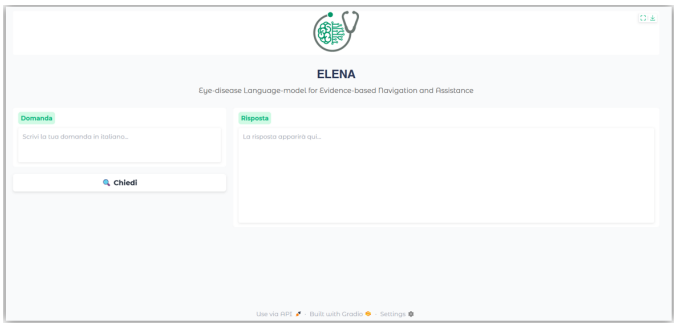


Fig. 2. ELENA framework and graphical user interface.

these measures, only reaches 0.117. These values suggest a modest agreement with the golden answers when evaluated through conventional automatic metrics. However, these numerical results must be interpreted with caution. The primary limitation lies in the "rigidity" of the metrics: they measure direct lexical or semantic overlap with the reference answers, without accounting for the human-like style, contextual reasoning, and explanatory enrichment that the LLM provides. Our model does not merely reproduce the content of the retrieved documents; it synthesizes and integrates information to deliver coherent, informative answers that often include additional relevant details about the diseases in question. For example, the LLM may highlight secondary symptoms, key risk factors, or typical disease progression that are not explicitly listed in the golden answers. Consequently, while the automated metrics report low similarity, the responses retain high practical and clinical value, offering an enriched, readable, and educational output that better aligns with the needs of healthcare professionals. Some example answers are reported in Table III, highlighting the summarizing and explanatory effectiveness of the framework. The table shows the question posed to the framework in English, its Italian counterpart, and the corresponding answer, again in English and Italian.

In this first implementation, the absence of many documents for some diseases could result in overly generic responses, as can be seen in case two reported in the table, or it may be due to the use of a non-optimized model. In addition, the translation-first approach carries intrinsic limitations. The extra translation steps introduce potential noise and may misinterpret nuanced medical terminology Consequently, further studies are needed to validate the framework under realistic operational conditions, where the document base is likely to be significantly broader and more diverse than in the current experimental setting.

In conclusion, it is important to emphasize that the primary goal of this work was to verify the feasibility of the proposed framework. With this in mind, the model was equipped with a simple graphical interface, allowing to easily test its response capabilities using the *Gradio* library. The interface is shown in Figure 2, which represents only a starting point.

V. CONCLUSION AND FUTURE DIRECTIONS

This study presented ELENA, a retrieval-augmented conversational framework designed to support clinicians and researchers in the ophthalmological rare disease domain. The results show that a modular hybrid approach-combining translation-based retrieval with an instruction-tuned LLM can generate medically coherent, context-aware answers within a lightweight, privacy-preserving setup suitable for small Italian healthcare institutions. Several important directions for future work must be taken in consideration. First, while we have demonstrated the feasibility of the framework, only a limited number of models and documents were tested, and the evaluation remains preliminary. A more comprehensive human evaluation—ideally conducted with young clinicians and researchers from small ASL centers will be essential to assess clinical relevance, readability and explainability, as previously demonstrated in [16] and [17]. Second, the dynamic knowledge base, continuously updated, should incorporate, ideally via secure ingest pipelines, versioning and audit-logging that respect patient privacy and institutional governance. Finally, to foster long-term adoption in small healthcare institutions, future work should focus on usability and integration: enhancing the graphical interface, deploying

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TABLE III
EXAMPLES OF OUTPUTS GENERATED BY THE MODEL (ITALIAN AND ENGLISH VERSION).

ID/Language	Question	Answer
1 - Italian	*Quali sono i sintomi principali della distrofia di Reis-Bücklers?*	*La distrofia di Reis-Bücklers è una rara distrofia autosomica dominante dello strato di Bowman, che si presenta nella prima infanzia con dolorose erosioni corneali ricorrenti, fotofobia e progressiva opacità corneale.*
2 - English	*What are the main symptoms of Reis-Bücklers dystrophy?*	*Reis-Bücklers corneal dystrophy is a rare autosomal dominant dystrophy of Bowman's layer, presenting in early childhood with painful recurrent corneal erosions, photophobia, and progressive corneal opacity.*
2 - Italian	*Quali segni oculari sono tipici della sindrome di Alport?*	*La sindrome di Alport è un disturbo ereditario del collagene di tipo IV che colpisce principalmente i reni, le orecchie e gli occhi. I sintomi comuni includono problemi di visione, danno renale e perdita dell'udito. Il trattamento è sintomatico e può comportare farmaci o interventi chirurgici per gestire i sintomi. Un medico può aiutare a diagnosticare la condizione basata su esami fisici e storia medica.*
2 - English	*What ocular signs are typical of Alport syndrome?*	*Alport syndrome is an inherited disorder of type IV collagen that primarily affects the kidneys, ears, and eyes. Common symptoms include vision problems, kidney damage, and hearing loss. Treatment is symptomatic and may include medications or surgical interventions to manage symptoms. A physician can help diagnose the condition based on physical examinations and medical history.*

the system in real clinical workflows, assessing user training needs and ensuring that the framework aligns with regulatory and ethical requirements in the healthcare domain. Following these directions would enable future work to progress from technical feasibility to a scalable, robust, and clinically viable knowledge-assistant system that strengthens the research and decision-support capacity of small Italian ASL institutions, while ensuring patient-data privacy and security.

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